

WORKSHEET - NAPLAN Year 9

Number and Algebra Solving Equations (9)

Teacher: Joanne Rust

Exam Equivalent Time: 10 minutes (based on NAPLAN allocation of ~1.25 min/mark)



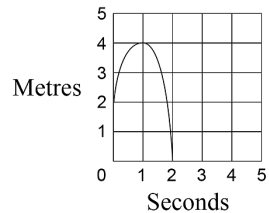
Questions

1. ☒ Number and Algebra, NAP-54357

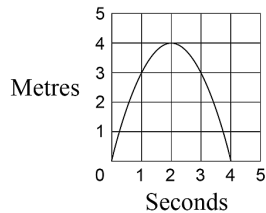
Thomas throws a ball from a height of 2 metres.

The ball reaches a maximum height of 4 metres and hits the ground again after 4 seconds.

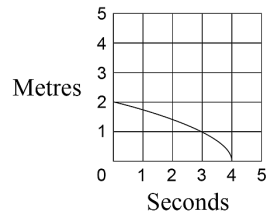
Which graph represents the path of Thomas' ball?



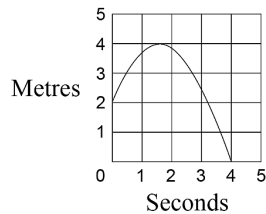
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2. ☒ Number and Algebra, NAP-30835

Ali joins a gym that has a fortnightly fee of \$33.

Members pay an extra \$12 per session if they want to do a special yoga class.

Which expression represents Ali's fortnightly gym bill if she does x yoga classes during the fortnight.

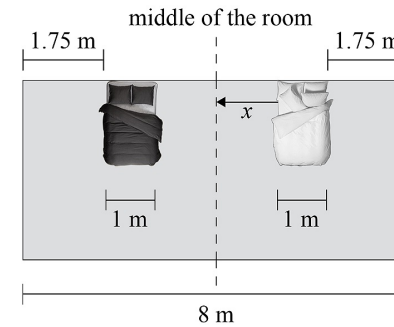
$12x$ $33x$ $33x + 12$ $33 + 12x$

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3. ☒ Number and Algebra, NAP-36779

Karen and Kim are sisters and share a bedroom.

They want their beds positioned the same distance from the centre of the room.



What is the distance x from the edge of each bed to the middle of the room?

1.25 m 2 m 2.25 m 2.75 m 5.25 m

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4. ☒ Number and Algebra, NAP-60301

Lily made a mobile phone call that lasted 39 minutes.

Her mobile phone provider charges her 19 cents per minute for calls.

Lily estimated that the total cost of the phone call was about \$8.

Lily's estimate was

☐ less than the actual cost.

☐ equal to the actual cost.

☐ more than the actual cost.

5. ✖ Number and Algebra, NAP-30867

Paul stacks milk cartons into supermarket refrigerator shelves.

Each shelf is stacked with 6 full cream milk cartons, 4 lite milk cartons and 2 skim milk cartons.

Every hour Paul stacks 240 milk cartons in total.

How many lite milk cartons does he stack every hour?

6. ✖ Number and Algebra, NAP-42723

Alice has 5 guinea pigs.

They eat 200 grams of cabbage per day, in total.

How many grams of cabbage would 1 guinea pig eat in one week?

40 grams 280 grams 1400 grams 2800 grams

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7. ✖ Numbers and Algebra, NAP-89766

It takes 2 people 10 hours to pick 800 lettuces during harvesting.

At the same rate, how many hours should it take for 5 people to pick 1200 lettuces?

hours

8. ✖ Number and Algebra, NAP-07375

Johnno is saving to buy a go-kart.

He created a table to keep track of his savings.

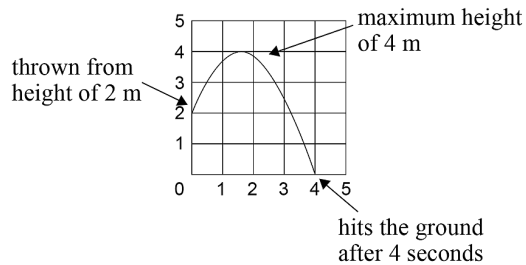
Number of months (x)	2	5	8	10
Amount of Johnno's savings account (y)	600	720	840	920

Complete the rule for the linear relationship between the amount of money in Johnno's savings account and the number of months of saving.

$y =$ $x + 520$

Worked Solutions

1. Number and Algebra, NAP-54357



2. Number and Algebra, NAP-30835

$$\begin{aligned}\text{Fortnightly cost} &= \text{fee} + \text{yoga classes} \\ &= 33 + 12x\end{aligned}$$

3. Number and Algebra, NAP-36779

Half the room is 4 m wide.

$$\begin{aligned}\therefore x &= 4 - (1 + 1.75) \\ &= 4 - 2.75 \\ &= 1.25 \text{ m}\end{aligned}$$

4. Number and Algebra, NAP-60301

An estimate of the cost could be just under:

$$\begin{aligned}40 \text{ minutes} \times 20 \text{ cents} &= \$8 \\ \therefore \$8 \text{ estimate is more than the actual cost.}\end{aligned}$$

5. Number and Algebra, NAP-30867

Shelves stacked each hour

$$= \frac{240}{12} = 20$$

$\therefore$ Lite milk cartons stacked

$$\begin{aligned}&= 20 \times 4 \\ &= 80 \text{ per hour.}\end{aligned}$$

6. Number and Algebra, NAP-42723

Cabbage eaten by 1 guinea pig in 1 day

$$\begin{aligned}&= \frac{200}{5} \\ &= 40 \text{ grams}\end{aligned}$$

$\therefore$ Cabbage eaten by 1 guinea pig in 1 week

$$\begin{aligned}&= 7 \times 40 \\ &= 280 \text{ grams}\end{aligned}$$

7. Numbers and Algebra, NAP-89766

It takes 20 hours to pick 800 lettuces.

$$\therefore 1 \text{ person can pick } \frac{800}{20} = 40 \text{ lettuces per hour}$$

$\therefore$ 5 people pick 200 lettuces per hour

$\therefore$ Time to pick 1200 lettuces (5 people)

$$\begin{aligned}&= \frac{1200}{200} \\ &= 6 \text{ hours}\end{aligned}$$

8. Number and Algebra, NAP-07375

After 2 months, \$600 in account

$$\begin{aligned}600 &= 2x + 520 \\ 2x &= 80 \\ x &= 40\end{aligned}$$

